Motion-Type-Aware Diagnostic Benchmarking and Targeted Correction of Video Frame Interpolation Models

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***Abstract*—Video Frame Interpolation (VFI) synthesizes intermediate frames between consecutive video frames and is widely used in slow-motion generation, frame-rate conversion, video restoration, and multimedia applications. Recent deep learning models such as RIFE, FILM, and IFRNet achieve strong performance on established benchmarks. However, aggregate evaluation using metrics such as Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and Learned Perceptual Image Patch Similarity (LPIPS) can conceal model-specific weaknesses under different motion conditions, including large or non-linear motion, occlusion, and non-rigid deformation.**

This work presents a motion-type-aware diagnostic benchmarking framework coupled with a lightweight targeted correction stage for pretrained VFI models. RIFE, FILM, and IFRNet are treated as frozen baselines and evaluated separately across four interpretable motion categories: near-linear motion, rotational or curved motion, occlusion-heavy scenes, and non-rigid deformation. Category-wise quality and efficiency measurements are used to construct failure profiles and identify the weakest motion condition for each model. A lightweight residual correction module is then trained specifically for the identified weak category and compared with both the unmodified baseline and a generic correction model trained across all motion categories. The implemented evaluation uses 3,601 paired clips across the four motion categories. Generic correction produces small-to-moderate average gains across the three baselines, while rotation-targeted correction does not consistently outperform generic correction on the 180 rotation clips. Measured inference times for the frozen baselines are 9.30 ms for RIFE, 11.03 ms for IFRNet, and 173.63 ms for FILM. The central contribution is therefore a diagnosis-to-correction methodology for interpretable evaluation of pretrained VFI models rather than the development of a new VFI backbone.

***Index Terms*—Video Frame Interpolation, Motion-Aware Diagnosis, Benchmarking, Failure Analysis, Lightweight Correction, RIFE, FILM, IFRNet**

I. Introduction

Video Frame Interpolation (VFI) aims to synthesize one or more frames between existing frames in a video sequence. Given two observed frames, an interpolation system estimates the appearance at an intermediate temporal position and generates a frame that is spatially plausible and temporally consistent with its neighbors. VFI is used in slow-motion generation, frame-rate up-conversion, video streaming, video restoration, animation, gaming, and immersive multimedia systems [1], [2]. The task is fundamentally difficult because the desired intermediate frame is unobserved, requiring a model to infer object motion, visibility changes, and image content that may be absent from either input frame.

The difficulty increases when the motion between input frames is not approximately linear. Large displacement can cause correspondence errors, while rotation and curved trajectories violate simple translational assumptions. Occlusion and disocclusion are especially challenging because a region may disappear from one frame and become visible in another, leaving incomplete evidence for direct reconstruction. Non-rigid objects such as people, cloth, animals, and fluid-like structures can also change their shape while moving. These conditions can produce artifacts such as ghosting, duplicated structures, blurry boundaries, and incorrect object geometry.

Early deep learning approaches investigated adaptive synthesis operations. Niklaus *et al.* proposed video frame interpolation using adaptive convolution, where spatially varying kernels are learned to synthesize the target frame [3]. Context-aware synthesis subsequently incorporated contextual features to improve reconstruction of challenging image regions [4]. Flow-based methods then became a major direction. Super SloMo estimated bidirectional optical flow and visibility maps to synthesize intermediate frames [1], while DAIN introduced depth-aware flow projection to improve handling of occlusion and depth ordering [2]. AdaCoF used adaptive collaboration of flows to provide multiple learned sampling locations for target pixels [5], and CAIN demonstrated that channel attention could support flow-free interpolation [6].

Recent models have increasingly emphasized efficiency and robust motion estimation. RIFE introduced IFNet for direct intermediate-flow estimation and a privileged distillation strategy, enabling fast interpolation without dependence on a separately pretrained optical-flow model [7]. FILM was designed for large motion and uses a shared multi-scale feature extractor together with a scale-agnostic motion estimator [8]. IFRNet uses an efficient encoder-decoder architecture that progressively refines bilateral intermediate flows and intermediate features [9]. These three models are useful baseline architectures because they represent established and publicly available VFI approaches with different design choices, and all three are used strictly as frozen, fixed pretrained checkpoints, without retraining or modification.

Transformer-based methods have further expanded VFI architectures. Video Frame Interpolation with Transformer uses cross-scale window attention to model long-range relationships

and improve large-motion handling [10]. Other approaches investigate flow-agnostic representations and multi-flow or splatting-based synthesis [11], [12]. Although these methods improve overall interpolation quality, their reported scores still depend strongly on the selected dataset, test protocol, and evaluation metrics. The present work does not add a Transformer-based or other new backbone; the existing pretrained RIFE, FILM, and IFRNet checkpoints remain the only VFI models under study.

A central issue motivating this project is therefore not the absence of strong VFI models, but the way their performance is interpreted. Conventional evaluation often summarizes a complete test set using average PSNR, SSIM, or perceptual scores. Such aggregate values are useful for ranking models but can hide motion-specific weaknesses. Two models may have similar average PSNR while exhibiting substantially different behavior on occlusion, rotation, or non-rigid deformation. A model that performs well on easy near-linear motion can statistically compensate for severe failures on a smaller difficult subset. Consequently, a single average score may not reveal the failure mode that is most important for practical use.

Recent benchmark research has explicitly identified weaknesses in VFI evaluation. Kiefhaber *et al.* proposed a dedicated benchmark with consistent evaluation and analysis with respect to per-pixel attributes such as motion magnitude [13]. LAVIB similarly introduced a large-scale benchmark in which clips are associated with measurable attributes including motion magnitude, luminance, sharpness, and contrast [14]. These studies support the broader idea that VFI performance should be interpreted in relation to input characteristics rather than only through one aggregate number. This study extends this direction toward semantically interpretable motion categories and an explicit link between diagnosis and a targeted correction step.

The contribution of this project is not a new VFI backbone. Instead, the project uses existing pretrained VFI architectures as controlled, frozen baselines and contributes an evaluation-and-correction methodology. First, a curated motion-type-tagged evaluation subset is constructed from existing datasets using a reproducible, automated tagging procedure based on independent optical-flow descriptors, rather than creating an entirely new large-scale dataset. Second, RIFE, FILM, and IFRNet are evaluated separately for each motion category using PSNR, SSIM, and LPIPS. Third, category-wise results are converted into failure profiles and used to identify model-specific weak conditions. Finally, lightweight residual correction modules are evaluated for the diagnosed conditions and compared with both the unmodified baseline and a generic (non-targeted) correction condition under predefined evaluation criteria. This makes the project an investigation of whether diagnostic benchmarking can guide an efficient, targeted improvement, rather than a claim that such improvement is guaranteed.

The research question is: *Can motion-type-aware diagnosis reveal weaknesses in pretrained VFI models that are hidden by aggregate metrics, and can a lightweight, category-specific correction module improve the identified weakness more effectively than a generic correction network, without substantially increasing computational cost?* The evaluation uses Vimeo-90K and DAVIS, with categories covering near-linear, rotational or curved, occlusion-heavy, and non-rigid motion [15], [16]. The study reports category-wise quality measurements, correction performance, and inference-time measurements obtained from the implemented pipeline.

II. Literature Survey

The literature on VFI can be broadly organized into kernel-based synthesis, flow-based interpolation, attention and Transformer-based models, efficient architectures, and recent benchmark-oriented research. Early work established that interpolation could be learned without explicitly constructing a conventional optical-flow pipeline. Niklaus *et al.* [3] solve per-pixel synthesis using spatially varying convolution kernels; the main strength of this approach is that it avoids explicit flow estimation entirely, but its limitation is poor handling of large displacements, since kernel receptive fields are spatially local. The context-aware extension [4] addresses part of this limitation by adding contextual features, improving reconstruction in regions where local appearance alone is insufficient, though large-motion and occlusion handling remain weak. These limitations motivate the shift toward explicit motion estimation studied in the present work's diagnostic categories.

Flow-based approaches made motion estimation explicit. Super SloMo [1] solves multi-frame interpolation using bidirectional flow and visibility maps; its strength is flexible arbitrary-time interpolation, but it assumes reasonably well-behaved flow estimation and does not explicitly diagnose failure by motion type. DAIN [2] improves occlusion and depth-ordering handling through depth-aware flow projection, directly addressing one of the four motion categories used in this work (occlusion-heavy motion), but does not generalize this idea into a systematic per-category evaluation protocol. AdaCoF [5] and CAIN [6] improve sampling flexibility and enable flow-free interpolation respectively, but both are evaluated only using aggregate metrics, which is precisely the evaluation gap this work targets.

Efficiency-oriented models are the three baselines adopted in this work. RIFE [7] solves the runtime cost of relying on a separately pretrained optical-flow network by estimating intermediate flow directly through IFNet with privileged distillation; its strength is speed and arbitrary interpolation time, while large or highly non-linear motion remains a documented weakness. FILM [8] specifically targets large-motion scenes using a shared multi-scale, scale-agnostic feature extractor; its strength is improved handling of large displacement, though performance under non-rigid deformation and occlusion is less thoroughly characterized. IFRNet [9] solves the efficiency-quality trade-off using a pyramid encoder-decoder with progressive bilateral flow refinement; it is competitive and efficient, but like RIFE and FILM, its published evaluation

relies on aggregate dataset-level metrics rather than motion-category-specific analysis. This shared limitation across all three baselines, aggregate-only evaluation, is the specific gap the proposed diagnostic framework addresses.

Transformer-based and flow-agnostic architectures have further diversified the design space. Lu *et al.* [10] use cross-scale window attention to enlarge the receptive field for large-motion handling; FLAVR [11] removes explicit flow estimation entirely; many-to-many splatting [12] represents complex motion using multiple motion vectors per pixel. These methods generally improve aggregate quality or efficiency but, similarly to the three chosen baselines, do not provide category-wise diagnostic evaluation. Because this work is deliberately scoped around existing baselines rather than new backbones, these architectures are discussed as context but are not adopted as baselines.

Several studies specifically target difficult motion. ST-MFNet [17] uses a spatio-temporal multi-flow architecture for large and complex motions and dynamic textures; its strength is handling dynamic, non-rigid content, but it does so through added architectural complexity rather than diagnosis of an existing model. H-VFI [18] addresses large motion through hierarchical, coarse-to-fine deformable kernels; the limitation, as with ST-MFNet, is that improvement is pursued only through new architecture design. Long-term VFI [19] investigates feature propagation for larger temporal gaps, extending the operating range of interpolation but not addressing motion-category-specific failure analysis. These works confirm that motion complexity is a major, recurring source of VFI failure, but their objective is architectural improvement rather than diagnosing already-trained models across interpretable motion categories, which is the specific direction pursued here.

Benchmarking research provides the most directly related perspective. Kiefhaber *et al.* [13] identify inconsistencies in existing VFI evaluation protocols and propose a benchmark with consistent error metrics and analysis with respect to per-pixel attributes such as motion magnitude; its strength is exposing evaluation inconsistency across prior work, but the benchmark stops at diagnosis and does not propose or evaluate any corrective mechanism. LAVIB [14] extends attribute-based benchmarking to a large scale, associating clips with motion magnitude, luminance, sharpness, and contrast; like Kiefhaber *et al.*, it provides diagnostic value but no corrective follow-up. Table I summarizes this comparison. The present study addresses this shared limitation by connecting diagnostic benchmarking with a corrective mechanism.

A. Recent Methods for Complex Motion and Reliability

Recent studies offer further context for motion-aware diagnosis and correction. Plack *et al.* introduce a Transformer-based interpolator that also predicts an uncertainty/error map, allowing unreliable regions to be identified and, in rendered scenes, selectively refined [20]. This is relevant to our diagnostic objective because it shows that estimating interpolation reliability can be useful beyond reporting a single image-quality score. Yu *et al.* propose range-nullspace synthesis with focalized motion estimation, emphasizing confidence-aware trajectory fitting and the suppression of unreliable flow candidates [21]. Xia *et al.* exploit high-temporal-resolution spike streams to improve interpolation in high-speed motion, occlusion, and motion-blur conditions [22]. These approaches address difficult motion through new inputs or architecture changes, rather than through post-hoc diagnosis of frozen models.

Generative methods have also become prominent. Jain *et al.* present VIDIM, which uses cascaded diffusion models to synthesize intermediate video content and is designed to handle ambiguous or nonlinear motion [23]. Huang *et al.* propose motion-aware latent diffusion (MADiff), incorporating motion priors during latent diffusion to improve temporal coherence under complex motion [24]. Liu *et al.* introduce direct synthesis using an event-based reference, targeting severe occlusion and cases where conventional flow estimation becomes unreliable [25]. Lew *et al.* propose disentangled motion modeling (MoMo), which applies diffusion to intermediate motion fields rather than directly modeling all pixel values [26]. Zhang *et al.* introduce EDEN, a diffusion-based approach for high-quality interpolation under large and nonlinear motion [27], while Hai *et al.* propose hierarchical flow diffusion to improve the efficiency of diffusion-based interpolation [28]. These methods indicate that complex motion remains a central challenge, but their primary goal is to improve interpolation generation rather than to compare failure profiles across unchanged pretrained models.

Evaluation methodology is also evolving. Daly *et al.* study computationally efficient motion-based quality metrics and propose a divergence-based metric that can complement conventional pixel metrics [29]. This supports reporting multiple complementary measures rather than relying exclusively on PSNR or SSIM. The present project differs from these methods by holding the three selected VFI baselines fixed, grouping clips into interpretable motion types, and testing whether diagnosis can guide a lightweight correction module. These methods are discussed as related work; the experimental comparison remains limited to the three selected pretrained baselines.

The proposed project is related to these benchmarking efforts but makes a different methodological choice. Instead of attempting to construct a new large-scale benchmark, it curates a smaller evaluation subset from established datasets and assigns motion-type labels using an automated, reproducible tagging procedure (Section II-C). The four categories are near-linear motion, rotational or curved motion, occlusion-heavy scenes, and non-rigid deformation. This categorization is intended to make the resulting failure analysis interpretable and directly connectable to a correction strategy. The benchmark is therefore a diagnostic subset rather than a replacement for established VFI datasets.

Recent surveys also indicate that VFI continues to face challenges involving large motion, occlusion, non-linear motion, lighting changes, and high-resolution processing [30], [31]. These recurring challenges motivate an evaluation methodol-

TABLE I

COMPARISON OF REPRESENTATIVE VFI AND BENCHMARKING WORKS AGAINST THE PROPOSED METHODOLOGY. "DIAGNOSTIC EVALUATION" INDICATES WHETHER THE WORK ANALYZES PERFORMANCE WITH RESPECT TO INPUT MOTION CHARACTERISTICS RATHER THAN ONLY AGGREGATE SCORES. "TARGETED CORRECTION" INDICATES WHETHER A DIAGNOSED WEAKNESS IS USED TO GUIDE A CORRECTIVE MECHANISM.

Work	**Main approach**	**Motion handling**	**Diagnostic evaluation**	**Targeted correction**
RIFE [7]	Direct intermediate-flow estimation (IFNet) with privileged distillation	Efficient, general motion; large/non-linear motion remains weak	No (aggregate only)	No
FILM [8]	Shared multi-scale, scale-agnostic feature extractor	Targets large motion specifically	No (aggregate only)	No
IFRNet [9]	Pyramid encoder-decoder, progressive bilateral flow refinement	Efficient general motion handling	No (aggregate only)	No
Kiefhaber *et al.* [13]	Consistent benchmark with per-pixel attribute analysis	Analyzes motion magnitude as an attribute	Yes	No
LAVIB [14]	Large-scale benchmark with attribute annotations	Motion magnitude, luminance, sharpness, contrast	Yes	No
Present study	Diagnostic benchmarking of frozen pretrained baselines + lightweight residual correction	Four explicit motion categories from optical-flow descriptors	Yes	Yes (model-specific, targeted)

ogy that asks not only which model has the highest average score, but also under which motion condition each model fails and by how much.

B. Baseline Architectures

The experimental study uses three established pretrained architectures: RIFE, FILM, and IFRNet, evaluated using fixed, publicly released pretrained checkpoints. They are not retrained or fine-tuned as part of the baseline evaluation, and their internal weights are frozen throughout the diagnostic stage. This provides a controlled comparison of existing VFI systems and keeps the project computationally feasible. For reproducibility, the implementation records, for each baseline: the exact checkpoint version and source, input resolution, requested interpolation time, preprocessing steps, evaluation hardware, and any inference-time settings (e.g., precision, batch size).

RIFE uses IFNet to estimate intermediate flows directly from the input frames and temporal information [7]. FILM uses a multi-scale feature extractor with shared weights and a scale-agnostic motion estimator to handle large scene motion [8]. IFRNet uses an efficient encoder-decoder structure with pyramid features and progressive refinement of bilateral intermediate flows and intermediate features [9]. These architectural differences make category-wise comparison scientifically useful, since each model is expected to respond differently to different motion conditions.

C. Motion-Type Tagging Methodology

To ensure that motion-category assignment is reproducible and free from subjective bias, all candidate clips are tagged automatically using dense optical flow computed by RAFT [32]. RAFT is used here strictly as an independent, frozen optical-flow estimator for motion characterization and category tagging; it is not one of the VFI models under evaluation, is not trained or fine-tuned as part of this work, and plays no role in frame synthesis. RIFE, FILM, and IFRNet remain the only VFI baselines being benchmarked, while RAFT's role is limited to supplying the motion descriptors described below.

For each candidate triplet (I_0, I_{gt}, I_1), four scalar descriptors are derived from the forward flow field F computed between I_0 and I_1. These are motion descriptors used for automated category assignment; they are not intended as a perfect semantic classifier of motion type, and category boundaries are treated as operational definitions rather than ground-truth labels.

- **Motion magnitude** m**:** mean flow magnitude $|F|$ over motion-active pixels. Motion-active pixels are defined as pixels whose flow magnitude exceeds a predefined threshold, providing a practical proxy for foreground motion without requiring an external semantic segmentation model.
- **Angular variance** a**:** circular variance of flow direction across neighboring pixels. For flow direction angles θ_i, the mean resultant length is

$$R = \sqrt{\left(\frac{1}{N}\sum_i \cos\theta_i\right)^2 + \left(\frac{1}{N}\sum_i \sin\theta_i\right)^2},$$

and the circular variance is defined as

$$a = 1 - R.$$

This formulation accounts for the circular nature of angular measurements and avoids discontinuities at the $0/2\pi$ boundary. Rotational or curved motion is characterized by high circular variance relative to the near-linear population, indicating that flow direction changes across the frame in a manner inconsistent with pure translation.
- **Occlusion ratio** o**:** proportion of motion-active pixels failing a forward–backward consistency check. Let F_{fwd} denote the forward flow and let F_{bwd} be sampled at

$x' = x + F_{\mathrm{fwd}}(x)$. A pixel is considered inconsistent when

$$\|F_{\mathrm{fwd}}(x) + F_{\mathrm{bwd}}(x')\| > \alpha\,(\|F_{\mathrm{fwd}}(x)\| + \|F_{\mathrm{bwd}}(x')\|) + \beta, \tag{1}$$

where α controls the relative tolerance and β provides an absolute tolerance floor. This adaptive relative–absolute threshold makes the consistency criterion less sensitive to the overall motion scale than a fixed threshold. Occlusion-heavy motion is characterized by a high occlusion ratio, indicating a large disoccluded or occluded area between I_0 and I_1.

- **Non-rigidity score** n: spatial variance of the local flow Jacobian (divergence and shear terms). Non-rigid deformation is characterized by high spatial variance in the local Jacobian, since rigid-body motion produces locally near-uniform Jacobians, whereas deforming content (cloth, water, fluid-like structures) produces spatially inconsistent local deformation.

The motion-active mask is used only for descriptor computation and does not represent semantic foreground segmentation. In the implementation, forward–backward consistency additionally excludes flow samples whose warped coordinates fall outside the image bounds, thereby preventing invalid samples from being treated as valid correspondences. For angular variance, pixels with near-zero flow magnitude are excluded so that static regions do not artificially increase the rotation score. For non-rigidity, both the divergence- and shear-based local-Jacobian variance and a Frobenius-norm Jacobian-inconsistency measure are computed and retained as candidate features for subsequent model selection.

Additional diagnostic features include the 90th-percentile flow magnitude, active-pixel ratio, mean forward–backward error, and mean relative forward–backward error.

Each descriptor is standardized (zero mean, unit variance) over the candidate clip pool. Rather than relying solely on a fixed percentile threshold and argmax rule, the implemented tagging pipeline evaluates multiple percentile settings and supervised classifiers using the manually labelled calibration subset. Decision Tree and Support Vector Machine (SVM) classifiers are evaluated using the standardized motion descriptors and additional diagnostic features. Four-fold cross-validation is used to compare classifier configurations and percentile settings. Feature-subset selection is also performed by evaluating combinations of the available diagnostic features. The final category assignment is produced by the selected classifier configuration, which is fixed before the final test evaluation. The four output categories are near-linear, rotational or curved, occlusion-heavy, and non-rigid deformation.

The calibration set contains 170 clips, including approximately 50 clips with manual motion-category labels used for supervised classifier calibration and evaluation. The remaining calibration clips are used for feature and threshold analysis without entering the final test set. All classifier and feature-selection decisions are fixed before final test evaluation to avoid test-set leakage.

The manually labelled subset provides reference categories for evaluating candidate tagging configurations. Classifier performance is assessed during calibration, while the final test clips remain unseen during classifier selection and correction-module training.

D. Dataset Construction and Splits

The primary datasets are Vimeo-90K [15] and DAVIS [16], both of which provide frame triplets or sequences from which a valid two-frame input and a ground-truth intermediate frame can be directly extracted (i.e., three temporally consecutive frames, with the first and third used as input and the second held out as ground truth). Because valid interpolation triplets require a genuine, temporally centered ground-truth frame, GoPro [33] is not treated as a VFI benchmark in the primary methodology, since it was constructed for dynamic scene deblurring and does not directly provide the same triplet structure; it is therefore excluded from the primary dataset pool. If additional samples are required for a specific category, this will be addressed through training-time augmentation applied to the primary datasets, not by introducing GoPro as an additional source of test triplets.

The candidate clip pool is split into three disjoint subsets prior to any category tagging or model training: a training set (used to fine-tune correction modules), a calibration set (including 170 calibration clips, of which approximately 50 have manual motion-category labels used for classifier calibration), and a test set (used only for final evaluation). Test samples are never used to train any correction network, and clips originating from the same source video are kept within a single split to reduce the risk of near-duplicate leakage between splits. The minimum number of tagged samples required per category for reliable correction-module training is predefined before final testing (rather than chosen after observing category populations); if a category falls below this predefined minimum in the training split, additional training examples are obtained only through synthetic augmentation (e.g., synthetic rotation or non-rigid warping applied to near-linear training clips), and this augmentation is applied exclusively to the training split. The test split is never augmented and always consists of real video samples, so that final evaluation reflects genuine motion conditions rather than synthetic ones.

E. Weakest-Category Selection Rule

For each baseline model, a diagnostic score is computed independently per motion category to enable a deterministic identification of the weakest category, without manual selection after inspecting results. For category c and model k, PSNR and SSIM are metrics for which higher values indicate better performance, while LPIPS is a metric for which lower values indicate better (more perceptually similar) performance. Each metric is min–max normalized across the four categories for a given model, so that within-model comparisons are on a common scale:

$$\widehat{X}_c = \frac{X_c - \min_{c'} X_{c'}}{\max_{c'} X_{c'} - \min_{c'} X_{c'}}, \tag{2}$$

where $X \in \{\text{PSNR}, \text{SSIM}\}$ (higher-is-better) or $X = \text{LPIPS}$ (in which case the normalized score is inverted, $\widehat{X}_c = 1 - \widehat{X}_c$, so that higher $\widehat{X}_c$ always denotes better performance across all three metrics). A single diagnostic score per category is then defined as a weighted combination

$$D_c = w_1 \widehat{\text{PSNR}}_c + w_2 \widehat{\text{SSIM}}_c + w_3 \widehat{\text{LPIPS}}_c, \tag{3}$$

with predefined, fixed weights (e.g., $w_1 = 0.5, w_2 = 0.25, w_3 = 0.25$, reflecting PSNR as the primary quality indicator and SSIM/LPIPS as supporting metrics). The weakest category for model k is then selected deterministically as

$$c_k^* = \arg\min_c D_c. \tag{4}$$

Because D_c is computed identically for all four categories using a fixed, predefined weighting before any correction module is trained, this procedure removes the possibility of manually selecting the weakest category after observing which category would be most convenient to correct.

F. Diagnostic and Correction Architecture

The implemented system does not replace the baseline architectures. Instead, it is organized as a two-stage evaluation and correction framework. In Stage 1, the original pretrained VFI model receives two input frames and produces the baseline interpolated frame I_b. This output is evaluated against the ground-truth intermediate frame I_{gt} using PSNR, SSIM, and LPIPS. Stage 2 is activated only after the diagnostic procedure in Section II-E identifies a weak motion category for that model.

As shown in Fig. 1, the lightweight correction module receives the two input frames and the baseline interpolated frame, concatenated as a nine-channel tensor. A shallow convolutional encoder extracts local features, followed by residual refinement blocks. Two output heads estimate a three-channel residual image R and a one-channel confidence mask M, the latter passed through a sigmoid activation to constrain its values to approximately $[0, 1]$. The corrected frame is computed as

$$I_c = I_b + M \odot R, \tag{5}$$

where $\odot$ denotes element-wise multiplication. The confidence mask M is used to control *where* the predicted residual is applied, allowing the network to leave regions it considers already accurate largely unchanged while concentrating correction in regions it identifies as erroneous; it is learned end-to-end as part of the correction network from training supervision and is not derived from ground-truth segmentation or obtained from any external source at inference time. It is not claimed to provide explicit semantic segmentation of the scene; it is only a learned, continuous per-pixel weighting.

The ideal (target) residual used to supervise training is defined as

$$R^* = I_{gt} - I_b, \tag{6}$$

i.e., the correction network is trained to estimate the per-pixel adjustment that moves the baseline interpolated frame toward the ground-truth intermediate frame. The ground-truth frame I_{gt} is used only during training and evaluation to compute this target and the corresponding losses; it is never available to, or used by, the correction network at inference time, at which point only I_0, I_1, and I_b are provided as input.

The implemented training objective is

$$L = L_1 + 0.1 L_{\text{Charbonnier}} + 0.001 L_{\text{res}}, \tag{7}$$

where L_1 is the mean absolute reconstruction error between I_c and I_{gt}, $L_{\text{Charbonnier}}$ is the Charbonnier reconstruction term, and L_{res} is a residual regularization term applied to the predicted correction residual. The weights are fixed to 1, 0.1, and 0.001, respectively.

G. Model-Specific Correction Strategy

Each baseline model (RIFE, FILM, IFRNet) is diagnosed independently using the procedure in Section II-E, following the pipeline Model $\rightarrow$ Motion Diagnosis $\rightarrow$ Model-Specific Weakest Category $\rightarrow$ Targeted Correction. Because the three baselines differ architecturally, they may exhibit different weakest motion categories, in which case up to three separate lightweight correction modules are trained, one per baseline, each targeting that specific model's identified weakness. If all three baselines are found to share the same weakest category, a single correction module is instead trained and evaluated across all three baselines as an additional test of cross-architecture generalization.

H. Experimental Conditions and Ablation Plan

To evaluate the effect of *targeted* correction, three experimental conditions are evaluated for each baseline model:

- **Condition A (Baseline):** the pretrained VFI model alone, with no correction module applied.
- **Condition B (Generic correction):** the baseline combined with a lightweight correction network of identical architecture, trained using samples drawn from all four motion categories rather than the weakest category alone.
- **Condition C (Targeted correction):** the baseline combined with a lightweight correction network trained only on the weakest motion category identified by the diagnostic procedure in Section II-E.

The comparison between Conditions B and C evaluates whether targeted, diagnosis-guided correction provides a measurable advantage over a generically trained refinement network, rather than any improvement being attributable only to the presence of a correction module.

Additional ablations are retained as future extensions of the implemented framework: (i) correction without the confidence mask; (ii) correction without the residual/skip connection; (iii) generic correction; and (iv) targeted correction. The results reported here focus on the baseline, generic-correction, and targeted-correction conditions.

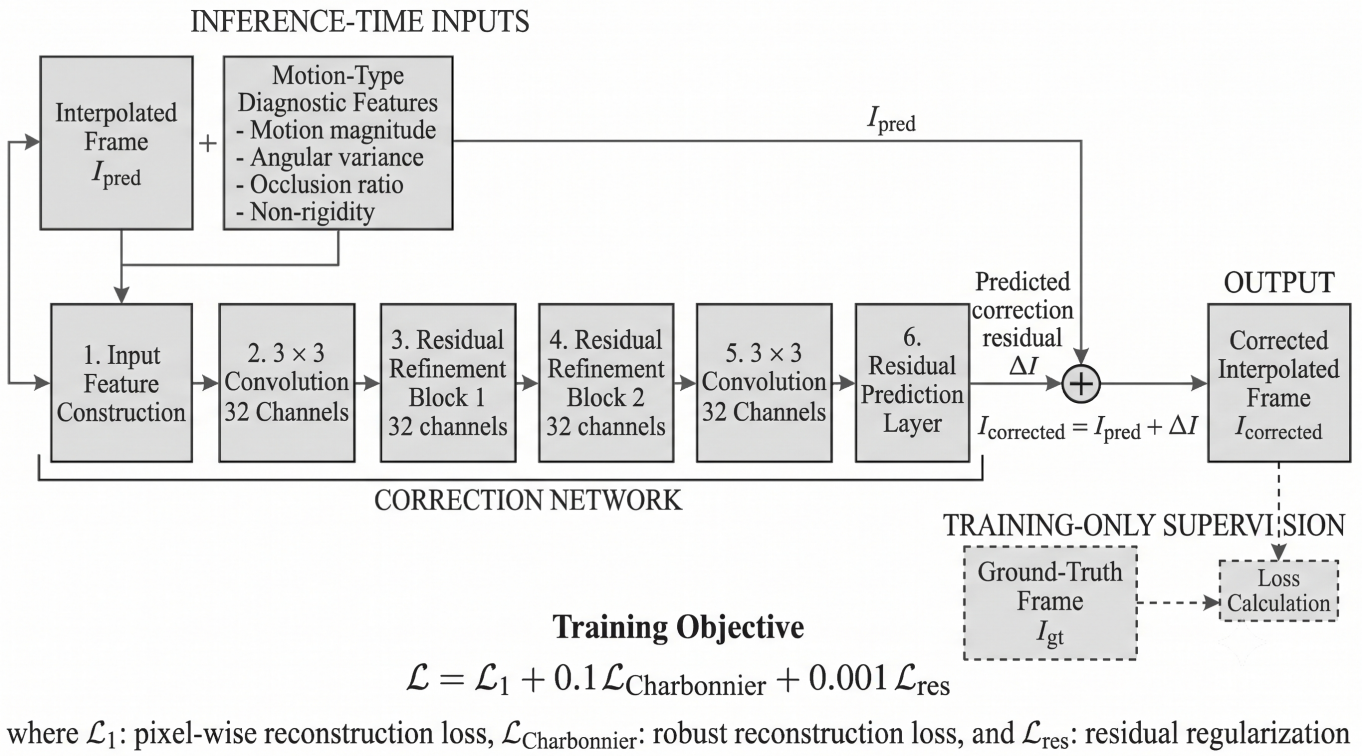


Fig. 1. Motion-type-aware VFI architecture. Existing pretrained VFI models (RIFE, FILM, and IFRNet) are used as frozen baselines. The diagnostic stage identifies the weakest motion category for each model, after which the proposed lightweight residual correction module is trained specifically for that category. The ground-truth intermediate frame is used only for evaluation and for computing the training target R^*, never at inference time.

I. Success Criteria

Success criteria are defined here as predefined evaluation targets against which the experimental results will be assessed; they are not presented as guaranteed or expected outcomes. The primary quality metric is PSNR. Targeted correction (Condition C) on a given baseline is considered to satisfy the primary success criterion if it achieves at least a 0.5 dB PSNR improvement over the baseline (Condition A) on the targeted weak category. SSIM improvement and/or LPIPS reduction on the same category are treated as supporting criteria corroborating the primary PSNR result, rather than independent pass/fail conditions. On non-target categories, degradation of more than 0.3 dB PSNR relative to the baseline is treated as an unacceptable side effect of correction. These numerical thresholds are predefined target criteria fixed in advance, not expected experimental results, and are set below typical improvement margins reported between successive VFI architectures in the literature [7], [8], so that satisfying them reflects a meaningful, non-trivial gain rather than measurement noise.

J. Efficiency Evaluation

The correction module is evaluated as a lightweight addition to the frozen VFI backbone using: (i) parameter count, (ii) floating-point operations (FLOPs) or an equivalent computational-cost measure, (iii) inference runtime, (iv) peak GPU memory usage, and (v) the percentage overhead of each of these relative to the corresponding frozen baseline model. Runtime and memory overhead are measured where available; the reported runtime results quantify the inference cost of the baseline models and provide the basis for assessing the practical overhead of correction. No additional efficiency claim is made beyond the measured inference-time results reported in the evaluation.

III. Operational Motion Categories

The four categories are designed to provide interpretable groupings for error analysis rather than to serve as perfect semantic descriptions of a scene. Near-linear motion is the operational reference category for clips not assigned to the other three categories by the classifier. Rotational or curved motion is associated with substantial variation in local flow direction. Occlusion-heavy motion is associated with elevated forward–backward flow inconsistency, which can arise when regions become hidden or newly visible between the two observed frames. Non-rigid motion represents clips assigned to the corresponding manually labelled class using the descriptor-based classifier.

The categories can overlap in physical scenes. A person running behind an object may exhibit non-rigid motion and occlusion simultaneously; a rotating camera may also produce spatially varying flow. The implemented classifier assigns a single dominant label to each clip, which makes category-wise tabulation straightforward but necessarily simplifies the

underlying scene. The reported test counts must therefore be interpreted as predicted operational categories rather than independently verified semantic ground truth for every test clip.

The near-linear group contains the largest number of clips, while rotation is the smallest group. This imbalance affects the interpretation of overall metrics: aggregate means are dominated by the more common categories. For this reason, the paper reports both overall changes and the direct rotation-subset comparison. In a future multi-label formulation, each clip could carry multiple motion attributes, allowing a clip to contribute to more than one diagnostic group. Such a change would require a different annotation and analysis protocol and is outside the present implementation.

IV. Optical-Flow Descriptor Rationale

RAFT is used as an independent flow estimator to characterize the input pair, not as an interpolation model. Using a shared estimator creates a common descriptor space for all three VFI baselines. If categories were derived from model-specific internal features, the meaning of a category could differ between models and make comparisons less consistent.

Angular variance summarizes variation in the direction of local flow vectors. A high value can be consistent with rotation or curved motion, but can also occur when several independently moving objects appear in the same frame. Mean forward–backward error measures disagreement between forward flow and backward flow after coordinate warping. Large disagreement can indicate occlusion or disocclusion, but can also result from motion blur, weak texture, large displacement, or optical-flow estimation error. Mean relative forward–backward error normalizes inconsistency by motion magnitude, reducing the extent to which the measure is determined solely by flow scale.

These descriptors are proxies. They do not directly measure object semantics, depth, or physical angular velocity. Their utility depends on whether the selected features correlate sufficiently with the manually assigned operational categories. Because RAFT is itself an estimator, its failures can propagate into the classifier. A stronger validation protocol would manually audit a held-out sample of test clips and report category-wise precision, recall, F1, and a confusion matrix. The available calibration cross-validation result should not be described as independent test-set classification performance.

V. Correction Network and Training Objective

The correction network receives a nine-channel tensor formed by concatenating the two observed RGB frames and the baseline RGB prediction:

$$X = \mathrm{Concat}(I_0, I_1, I_b). \tag{7}$$

Its outputs are a three-channel residual R and a one-channel confidence gate M. A sigmoid operation constrains the gate to $[0, 1]$. The residual formulation is intended to adjust the baseline prediction rather than synthesize an entirely independent frame. A small gate value suppresses the predicted residual at a location, while a larger value permits more of the residual to be added.

The implemented objective is

$$\mathcal{L} = \mathcal{L}_1 + 0.1\mathcal{L}_{\mathrm{Charbonnier}} + 0.001\mathcal{L}_{\mathrm{res}}, \tag{8}$$

where the first term is the mean absolute reconstruction error, the second is a robust Charbonnier reconstruction term, and the final term penalizes residual magnitude. The supplied training configuration uses five epochs and a learning rate of 10^{-4}. These settings describe the implemented experiment; they should not be interpreted as the result of an exhaustive hyperparameter search unless such a search was separately conducted.

Residual learning has a practical advantage: the correction network can learn to make spatially varying adjustments to the baseline output. However, the architecture alone does not guarantee improvement. If the training distribution does not represent test-time errors, the residual can be ineffective or can introduce artifacts. The confidence gate is also not automatically a calibrated probability of correctness; it is a learned modulation mask optimized through the training objective. Its values should not be interpreted probabilistically without a separate calibration experiment.

VI. Evaluation Metrics and Reporting Conventions

PSNR is derived from mean squared reconstruction error:

$$\mathrm{PSNR} = 10 \log_{10} \left(\frac{L^2}{\mathrm{MSE}} \right), \tag{9}$$

where L is the maximum image intensity. Larger PSNR indicates lower pixel-wise error. SSIM compares local luminance, contrast, and structure, with larger values generally preferred. LPIPS compares deep feature representations and is treated as a perceptual distance, for which smaller values are preferred. These metrics capture different aspects of reconstruction and should be reported separately.

For PSNR and SSIM, the reported change is treatment minus baseline. For LPIPS, the overall table reports baseline LPIPS minus treatment LPIPS, so a positive LPIPS reduction indicates improvement. In the rotation-targeted-versus-generic table, the reported differences are targeted minus generic for all columns; therefore, negative LPIPS differences favour the targeted condition. Explicitly defining these conventions avoids reversing the interpretation of the perceptual metric.

Paired measurements are useful because each treatment is evaluated on the same test clips. However, the mean change does not show how broadly the change is distributed across clips. A small positive mean could arise from modest improvements on most clips or from large gains on a few clips combined with neutral or negative changes elsewhere. Confidence intervals, per-clip distributions, and the proportion of clips improved would provide a more complete statistical account. Those quantities are not inferred from the aggregate values currently available.

VII. Experimental Procedure and Reproducibility

The evaluation pipeline consists of the following stages:

1) Load the fixed training, calibration, and test split manifests and verify that the partitions do not overlap.
2) Extract RAFT-based descriptors from the two observed input frames.
3) Fit the motion classifier using manually labelled calibration examples.
4) Predict an operational motion category for each test clip.
5) Generate a baseline intermediate frame using each frozen VFI checkpoint.
6) Generate corrected outputs under the generic and rotation-targeted conditions.
7) Compute PSNR, SSIM, and LPIPS against the same ground-truth middle frame.
8) Aggregate paired changes overall and for the predicted rotation category.
9) Measure baseline inference time separately from the corrected end-to-end pipeline.

The ground-truth middle frame must not be used to assign test categories or as an input to the correction network at inference time. It is used only as a training target and evaluation reference. This distinction prevents target-frame information from leaking into the diagnostic or prediction stage.

Reproducibility also depends on checkpoint identity, image normalization, resizing, color-channel order, image range, precision mode, batch size, warm-up iterations, and metric-library settings. These values should be taken from the actual notebooks and execution logs rather than guessed. The final runtime table contains baseline inference measurements; it does not provide the end-to-end latency of the corrected pipeline. Consequently, the paper does not claim that correction overhead has been demonstrated to be negligible.

VIII. Results and Discussion

The paired evaluation contains 3,601 clips: 1,928 near-linear (53.5%), 1,171 non-rigid (32.5%), 322 occlusion-heavy (8.9%), and 180 rotational (5.0%). All treatment results are paired with the corresponding frozen-model baseline; higher PSNR and SSIM and lower LPIPS indicate improvement.

Table II shows that the generic correction condition improves PSNR, SSIM, and LPIPS for all three baselines on the paired evaluation population. The magnitude of the gain differs substantially by model. Rotation-targeted correction also produces positive average SSIM and LPIPS changes for all three models, but its PSNR change is small for FILM and slightly negative for IFRNet. These aggregate results do not by themselves establish a rotation-specific advantage, so the targeted and generic conditions are compared directly on the 180 rotation clips.

For RIFE and IFRNet, the generic condition has higher PSNR and SSIM and lower LPIPS than the rotation-targeted condition on the rotation subset. For FILM, the two conditions are nearly identical in PSNR and SSIM, while the targeted condition has a small LPIPS reduction. Thus, within this evaluation, targeted correction does not provide a consistent advantage over generic correction on rotation clips. This finding is important because it tests the diagnosis-guided hypothesis directly rather than treating any improvement from adding a correction module as evidence of targeting benefit.

The measured runtimes in Table IV show substantially different computational costs among the frozen baselines. These values provide the reference point for evaluating the additional cost of the lightweight correction stage. Parameter-count and memory-overhead measurements are not used to make an unsupported efficiency claim and are left for subsequent reporting if available.

The predefined 0.5 dB PSNR target remains a criterion for a successful targeted improvement on the diagnosed weak category. The reported overall targeted changes are well below this threshold; consequently, the current results should be interpreted as evidence that the correction module can modify interpolation quality, but not as evidence that the targeted strategy has met the predefined improvement target.

IX. Threats to Validity

A. Internal Validity

Internal validity concerns whether observed changes can be attributed to the correction condition rather than unrelated implementation differences. Evaluating treatments on the same test clips supports paired comparisons, but does not guarantee identical preprocessing, checkpoints, or metric settings. All outputs should be traceable to the exact baseline checkpoint and input transformation used for each condition. The FILM rotation-targeted values require particular provenance verification against the intended checkpoint and final evaluation output before publication.

Training randomness may also affect results. Initialization, data order, augmentation, and nondeterministic GPU operations can cause variation between correction runs. The reported tables summarize the available final outputs and do not establish stability across repeated seeds. A stronger follow-up would train each correction condition several times using fixed, documented seeds and report variability across runs. Until then, the findings should be treated as descriptive results for the implemented configuration.

B. Construct Validity

Construct validity concerns whether the chosen measurements represent the concepts being discussed. PSNR measures pixel-wise reconstruction error, SSIM emphasizes local structural similarity, and LPIPS compares deep feature representations. None fully captures temporal consistency over a long sequence or human preference. Since the study evaluates an intermediate frame from a triplet, its conclusions should be limited to frame-level quality under the reported metrics.

The motion categories are operational constructs. Angular variance is a proxy for variation in flow direction rather than a direct semantic label for camera or object rotation. Forward–backward inconsistency can indicate occlusion, but it can also reflect flow-estimation errors. Non-rigid motion

TABLE II
OVERALL CHANGE FROM THE CORRESPONDING BASELINE OVER THE 3,601 PAIRED CLIPS.

Model	Condition	ΔPSNR (dB)	ΔSSIM	LPIPS reduction
RIFE	Generic	+0.1427	+0.001256	0.000045
RIFE	Rotation-targeted	+0.0322	+0.000227	0.000150
FILM	Generic	+0.0099	+0.000063	0.000070
FILM	Rotation-targeted	+0.0070	+0.000054	0.000108
IFRNet	Generic	+0.0204	+0.000298	0.000122
IFRNet	Rotation-targeted	-0.0003	+0.000013	0.000062

TABLE III
ROTATION-TARGETED VERSUS GENERIC CORRECTION ON THE 180 ROTATION CLIPS.

Model	ΔPSNR (dB)	ΔSSIM	ΔLPIPS
RIFE	-0.0582	-0.000822	+0.000106
FILM	-0.0026	-0.000004	-0.000036
IFRNet	-0.0113	-0.000228	+0.000112

TABLE IV
MEASURED BASELINE INFERENCE TIME.

Model	Mean (ms)	Median (ms)	P95 (ms)	FPS
RIFE	9.30	9.06	10.38	107.5
IFRNet	11.03	10.75	13.17	90.7
FILM	173.63	168.64	170.45	5.8

may overlap with camera motion and multiple independently moving objects. These limitations constrain the interpretation of category-wise results but do not remove the usefulness of a consistent diagnostic grouping.

C. External Validity

The experiments use a fixed Vimeo-90K-based test split and the supplied pretrained checkpoints. Performance may differ on other datasets, video resolutions, frame intervals, or real-world capture conditions. The results should not be generalized automatically to every VFI model or deployment scenario. Inference times depend on the GPU, precision mode, implementation, and timing protocol; they are not universal throughput guarantees.

Additional evaluation on a second independently labelled dataset would help determine whether the observed motion-specific behaviour transfers beyond the current test distribution. Such a study should preserve the same category definitions and metric conventions where possible, while acknowledging differences in dataset composition. The current manuscript does not claim cross-dataset generalization where such measurements are unavailable.

D. Statistical Conclusion Validity

The available summary reports mean changes but does not provide per-clip confidence intervals, statistical tests, or repeated training runs. Small differences, including the near-zero IFRNet PSNR change under rotation-targeted correction, may be sensitive to sample composition or training configuration. A future analysis should calculate paired per-clip differences and bootstrap confidence intervals separately for each model, condition, category, and metric.

A positive mean does not establish that most clips improved, and a negative mean does not imply that every clip degraded. Reporting the median paired difference and fraction of clips improved would expose this distribution. Where multiple models, categories, and metrics are tested, the analysis should state whether any correction for multiple comparisons is used.

X. PRACTICAL IMPLICATIONS

Motion-aware evaluation can support model selection when an application has a known distribution of motion types. A system intended for sports footage may place greater emphasis on rapid movement and occlusion, while animation or character footage may include substantial deformation. Aggregate quality remains important, but category-wise profiles can reveal whether a model's weaknesses align with the content expected in deployment.

The correction results show why diagnosis-guided interventions need direct controls. A targeted model may specialize in a selected category, but it also sees a narrower distribution of training examples and can be more sensitive to category-label noise. Generic correction can learn broader residual patterns and may therefore generalize better to a predicted target category. The present measurements are consistent with that possibility, but do not establish a causal explanation for the observed differences.

Efficiency must be considered jointly with quality. A correction stage that gives a small metric gain but adds considerable latency may be unsuitable for interactive use, while a modest gain with very low overhead could be useful for offline restoration. The measured baseline runtimes provide a reference point, but end-to-end correction latency, peak memory, parameter count, and FLOPs must be measured before making a deployment-level lightweight claim.

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